

LOCALIZED BOST–CONNES SYSTEMS III: INVARIANTS, ENTROPY, AND THE MEASURE CLASS

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ABSTRACT. The obstruction group $\Xi_\beta(K, S)$ of [13] is a complete invariant of the phase structure of the localized Bost–Connes system. We determine what kind of invariant it is, and the answer organizes everything else: Ξ_β *is an invariant of a measure class, not of a C^* -algebra*. It is the kernel of the natural map from $\widehat{G}_S$ to the *measurable* first cohomology of the tail equivalence relation taken with respect to π_β , and the whole of its dependence on β sits in that measure class.

Three consequences follow, and they explain a series of otherwise puzzling vanishings. First, at the level of K -theory the KMS simplex is exactly the set of states ω on the pointed ordered group $K_0(C(Y_S)) = C(Y_S, \mathbb{Z})$ satisfying the eigenvalue equations $\omega \circ \alpha_{\mathfrak{p}*} = N\mathfrak{p}^{-\beta}\omega$; and since the image of $K_0(C(Y_S))$ in $K_0(\mathcal{A}_{K,S})$ is the group of α_* -coinvariants, the passage to the crossed product annihilates precisely those eigenvectors. Second, every homotopy-invariant construction is blind: Connes–Thom gives $K_0(\mathcal{B}_{K,S}) = 0$ for the continuous core, so every dual-trace pairing on it vanishes identically; the dual flow acts trivially on $K_*(\mathcal{B}_{K,S})$ because $\mathbb{R}$ is connected; and the groupoid homology of $\mathcal{G}_S$, with or without twists by the modular cocycle, carries no β at all. Third, no dynamical entropy is available: $\alpha_{\mathfrak{a}}$ and the transfer operator $\mathcal{L}_{\mathfrak{a}}$ preserve no KMS_β state and indeed no invariant probability measure exists, in accordance with the type III classification; Voiculescu’s topological entropy of $\alpha_{\mathfrak{a}}$ vanishes because multiplication by $\mathfrak{a}$ is an ultrametric contraction; and the modular flow, which does preserve every KMS state, has zero CNT entropy because it lies in the compact gauge group.

What does see the phase structure is relative. We prove $S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}}) = |A_\beta| \log 2$, exactly $\log 2$ per threshold crossed and $+\infty$ once infinitely many have been, and show that the entire value is carried by the centre, so the type III factors contribute nothing. The resulting staircase coincides with the subfactor index staircase of Paper II of the series. We close by correcting the thermodynamic reading: all KMS_β states at a given β are equilibrium states on an equal footing, so the condensation of a phase bit is not a Landauer erasure.

Along the way we fix two structural boundaries that are easy to misplace. A KMS_β state with $\beta \neq 0$ is not a trace, so it defines no functional on $K_0(\mathcal{A}_{K,S})$; the K -theoretic locus of the KMS states is the eigenvalue problem on $K_0(C(Y_S))$ instead. And the de la Harpe–Skandalis determinant is natively defined on $\mathcal{U}_0$, which is the subgroup by which $K_1 = \mathcal{U}/\mathcal{U}_0$ is the quotient, so it induces no invariant on K_1 .

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Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/III-invariants/](#).

1. INTRODUCTION: WHICH KIND OF INVARIANT

We use the notation of [13]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ with $\sigma_t(\mu_a) = N a^{it} \mu_a$, symmetry group G_S , and

$$(1.1) \quad \Xi_\beta(K, S) = \left\{ \chi \in \widehat{G_S} : \sum_{\mathfrak{p} \in S} N \mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2 < \infty \right\},$$

the KMS_β simplex being $\text{Prob}(G_S/\Xi_\beta^\perp)$. Write $\mathcal{R}_S$ for the tail equivalence relation on $V = \mathbb{Z}_{\geq 0}^S$ and π_β for the forced product measure of [13, Lem. 2.10], $\alpha_{\mathfrak{p}}$ for the endomorphism $f \mapsto \mu_{\mathfrak{p}} f \mu_{\mathfrak{p}}^*$ of $C(Y_S)$, and $\mathcal{B}_{K,S} = \mathcal{A}_{K,S} \rtimes_{\sigma} \mathbb{R}$ for the continuous core.

The series. The present paper is III in the series. The foundational paper is [13], which classifies the KMS states and introduces Ξ_β ; Paper I [12] treats factor type and the spectrum of transitions. Papers II (ideal structure, boundary quotient, index), IV (noncommutative metric geometry) and V (the free variant) are in preparation, and nothing here depends on them; we refer to them by numeral, without citation.

The organizing statement is Theorem 2.3: Ξ_β is the kernel of a map into a *measurable* cohomology group computed relative to π_β . Since a C^* -algebra, a groupoid and a homotopy functor all forget a measure class, one should expect them to forget Ξ_β , and §§3–5 confirm this in each case. The point of collecting the vanishings is not that they are surprising individually — several are easy — but that they have a single cause, and that the cause identifies where the invariant does live.

2. Ξ_β IS MEASURABLE FIRST COHOMOLOGY

Let $\sigma : \mathbb{Z}^{(S)} \rightarrow G_S$ be the canonical homomorphism, regarded as a cocycle on $\mathcal{R}_S$ via $\sigma(v + a, v) = \sigma_a$.

Lemma 2.1 (Coboundary dictionary). *Let A be a second countable locally compact abelian group, $\Phi : \mathbb{Z}^{(S)} \rightarrow A$ a homomorphism, $\psi \in \widehat{A}$, and $c_{\mathfrak{p}} = \overline{\psi(\Phi(e_{\mathfrak{p}}))} \in \mathbb{T}$. The following are equivalent.*

- (1) $\psi \circ \Phi$ is a coboundary on $(V, \pi_\beta, \mathcal{R}_S)$: there is measurable $g : V \rightarrow \mathbb{T}$ with $g(v + e_{\mathfrak{p}}) = c_{\mathfrak{p}} g(v)$ a.e., for every $\mathfrak{p}$;
- (2) $\sum_{\mathfrak{p}} (1 - |m_{\mathfrak{p}}|) < \infty$, where $m_{\mathfrak{p}} = (1 - t_{\mathfrak{p}})/(1 - c_{\mathfrak{p}} t_{\mathfrak{p}})$ and $t_{\mathfrak{p}} = N \mathfrak{p}^{-\beta}$;
- (3) $\sum_{\mathfrak{p}} N \mathfrak{p}^{-\beta} |1 - \psi(\Phi(e_{\mathfrak{p}}))|^2 < \infty$.

Moreover, if these hold, then ψ is trivial on the group $E(\Phi)$ of essential values of Φ ; so $E(\Phi) \subseteq \Lambda^\perp$ with $\Lambda = \{\psi : \psi \circ \Phi \text{ a coboundary}\}$.

Proof. (1) $\Leftrightarrow$ (2) is Propositions 3.3 and 3.4 of [13]; those proofs use only that $c_{\mathfrak{p}} \in \mathbb{T}$, that the coordinates $v_{\mathfrak{p}}$ are independent under π_β with $\mu_{\mathfrak{p}}(k) = (1 - t_{\mathfrak{p}}) t_{\mathfrak{p}}^k$, and that $|g| = 1$. The target group of ψ plays no role. (2) $\Leftrightarrow$ (3) is [13, Lem. 3.2] together with $|1 - c_{\mathfrak{p}}|^2 = 2(1 - \text{Re } c_{\mathfrak{p}})$ and $|1 - c_{\mathfrak{p}}| = |1 - \overline{c_{\mathfrak{p}}}|$. For the last statement, a coboundary has no essential value other than 1, and ψ maps essential values of Φ to essential values of $\psi \circ \Phi$ [10]; so $\psi(E(\Phi)) = \{1\}$. $\square$

Remark 2.2. The converse, that ψ trivial on $E(\Phi)$ makes $\psi \circ \Phi$ a coboundary, holds when Φ is regular in the sense of Schmidt [10], i.e. cohomologous to a cocycle with values in $E(\Phi)$, and then $E(\Phi) = \Lambda^\perp$ by Pontryagin duality. For $A = G_S$ compact this is automatic, and Theorem 2.3 needs only (1) $\Leftrightarrow$ (3). For the cocycle with values in $G_S \times \mathbb{R}$ that governs the factor type, regularity is not known, and [12, Prop. 2.2 and Rem. 2.3] obtains from the inclusion $E(\Phi) \subseteq \Lambda^\perp$ an upper bound on the type only.

Theorem 2.3 (Cohomological description). *For every $\beta > 0$,*

$$(2.1) \quad \Xi_\beta(K, S) = \ker \left(\widehat{G_S} \xrightarrow{\chi \mapsto [\chi \circ \sigma]} H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T}) \right),$$

measurable 1-cohomology relative to the measure class of π_β , in the sense of Feldman–Moore [7].

Proof. Apply Lemma 2.1 with $A = G_S$, $\Phi = \sigma$: condition (3) is (1.1), and (1) says the class of $\chi \circ \sigma$ vanishes in H_{meas}^1 . $\square$

Corollary 2.4 (Where the β -dependence sits). *In (2.1) the objects $\widehat{G}_S$, $\mathcal{R}_S$ and σ are β -independent. The entire dependence on β is carried by the measure class of π_β , through which H_{meas}^1 is computed.*

Remark 2.5. Corollary 2.4 is the reason for everything below. K -theory is an invariant of a C^* -algebra; groupoid homology is an invariant of a topological groupoid; dynamical entropy is an invariant of a measure-preserving map. None of the three retains a measure class that is merely quasi-invariant. The vanishings of §§3–5 are therefore not failures of ingenuity but category errors, and we record them in that spirit.

3. K -THEORY: THE EIGENVALUE EQUATION

We first correct a statement announced in an earlier draft.

Proposition 3.1 (KMS states do not act on $K_0(\mathcal{A}_{K,S})$). *Let φ be a KMS_β state with $\beta > 0$ and $S \neq \emptyset$. Then $[p] \mapsto \varphi(p)$ does not define a homomorphism $K_0(\mathcal{A}_{K,S}) \rightarrow \mathbb{R}$.*

Proof. A positive functional descends to K_0 only if it is constant on Murray–von Neumann classes, i.e. only if $\varphi(vv^*) = \varphi(v^*v)$ for all v , i.e. only if φ is a trace. Take $v = \mu_{\mathbf{p}}$: then $v^*v = 1$ and $vv^* = \mathbf{1}_{\mathbf{p}Y_S}$, so $[1] = [\mathbf{1}_{\mathbf{p}Y_S}]$ in K_0 , while the KMS condition gives $\varphi(1) = 1$ and $\varphi(\mathbf{1}_{\mathbf{p}Y_S}) = N\mathbf{p}^{-\beta} < 1$. $\square$

The remedy is to work on the coefficient algebra, where a KMS state restricts to a measure and hence to a trace.

Theorem 3.2 (The KMS simplex as an eigenvalue problem). *Let $\alpha_{\mathbf{p}*}$ be the endomorphism $\mathbf{1}_U \mapsto \mathbf{1}_{\mathbf{p}U}$ of the pointed ordered group $(K_0(C(Y_S)), K_0^+, [1]) = (C(Y_S, \mathbb{Z}), C(Y_S, \mathbb{Z})_+, 1)$. Then $\varphi \mapsto \omega_\varphi$, $\omega_\varphi[\mathbf{1}_U] := \varphi(\mathbf{1}_U)$, is an affine bijection*

$$\text{KMS}_\beta(\mathcal{A}_{K,S}) \xrightarrow{\sim} \left\{ \omega \text{ a state of } (C(Y_S, \mathbb{Z}), C(Y_S, \mathbb{Z})_+, 1) : \omega \circ \alpha_{\mathbf{p}*} = N\mathbf{p}^{-\beta} \omega \quad \forall \mathbf{p} \in S \right\}.$$

Proof. By [13, Prop. 2.8] a KMS_β state is $\nu \circ E$ for a unique scaling-invariant ν , and ν is a Borel probability measure on the compact totally disconnected space Y_S , hence determined by its values on clopen sets; conversely a positive normalized finitely additive assignment on the clopen algebra extends to a regular Borel measure. Since $C(Y_S)$ is commutative, ν is a trace and ν_* is a well-defined state of the pointed ordered group. The scaling relation $\nu(\mathbf{a}E) = N\mathbf{a}^{-\beta}\nu(E)$ reads on clopen sets as $\omega(\alpha_{\mathbf{p}*}[\mathbf{1}_U]) = N\mathbf{p}^{-\beta}\omega[\mathbf{1}_U]$, and the relations for $\mathbf{p} \in S$ generate those for all $\mathbf{a} \in J_S$. $\square$

Corollary 3.3 (The crossed product annihilates the datum). *In $K_0(\mathcal{A}_{K,S})$ one has $\iota_*[\mathbf{1}_U] = \iota_*[\mathbf{1}_{\mathbf{p}U}]$ for every clopen $U \subseteq Y_S$ and $\mathbf{p} \in S$, since $\mu_{\mathbf{p}}\mathbf{1}_U$ is a partial isometry from $\mathbf{1}_U$ to $\mathbf{1}_{\mathbf{p}U}$. So $\iota_* : C(Y_S, \mathbb{Z}) \rightarrow K_0(\mathcal{A}_{K,S})$ factors through the group of coinvariants $C(Y_S, \mathbb{Z}) / \sum_{\mathbf{p}} (1 - \alpha_{\mathbf{p}*})C(Y_S, \mathbb{Z})$, and a functional on $C(Y_S, \mathbb{Z})$ that factors through $K_0(\mathcal{A}_{K,S})$ satisfies $\omega \circ \alpha_{\mathbf{p}*} = \omega$ for every $\mathbf{p}$. By Theorem 3.2 the KMS_β functionals satisfy instead $\omega \circ \alpha_{\mathbf{p}*} = N\mathbf{p}^{-\beta}\omega$ with $N\mathbf{p}^{-\beta} \neq 1$ for $\beta \neq 0$. Hence no nonzero KMS_β functional factors through $K_0(\mathcal{A}_{K,S})$: passing from the coefficient algebra to the crossed product is exactly the quotient that kills the eigenvectors of interest. (For a single prime the image of ι_* is the whole cokernel $\text{coker}(1 - \alpha_*) = K_0(\mathcal{A}_{K,S})$ and $K_1 = 0$; for several primes the image is the bottom piece of a filtration of K_0 by Koszul homology, computed in Paper II of the series.)*

Remark 3.4. Theorem 3.2 is a statement about the pair $(K_0(C(Y_S)), \alpha_*)$ together with a real parameter, not about a single abelian group. Every functor that forgets α_* , or that forces it to act as the identity, loses the arithmetic; and K_0 of the crossed product does the latter by construction.

4. HOMOTOPY INVARIANTS ARE BLIND

Theorem 4.1 (The continuous core). *For every K , S and $\beta > 0$:*

- (1) $K_0(\mathcal{B}_{K,S}) \cong K_1(\mathcal{A}_{K,S})$ and $K_1(\mathcal{B}_{K,S}) \cong K_0(\mathcal{A}_{K,S})$;
- (2) the dual flow $\hat{\sigma}$ induces the identity on $K_*(\mathcal{B}_{K,S})$;
- (3) the dual weight $\hat{\varphi}$ of any KMS_β state, a faithful semifinite trace on $\mathcal{B}_{K,S}$, vanishes on every projection in its domain (in every matrix algebra over $\mathcal{B}_{K,S}$); in particular the induced map $\hat{\varphi}_*$ on K_0 of its domain ideal is zero, and the pairing of $\hat{\varphi}$ with $K_0(\mathcal{B}_{K,S})$ carries no information.

Consequently no invariant built from $(K_*(\mathcal{B}_{K,S}), \hat{\sigma}_*, \text{trace pairings})$ distinguishes systems with different Ξ_β .

Proof. (1) is the Connes–Thom isomorphism $K_i(\mathcal{A} \rtimes_\sigma \mathbb{R}) \cong K_{i+1}(\mathcal{A})$ [4].

(2) The map $s \mapsto \hat{\sigma}_s$ is a point-norm continuous path of $*$ -automorphisms with $\hat{\sigma}_0 = \text{id}$, hence a homotopy of $*$ -homomorphisms $\mathcal{B} \rightarrow \mathcal{B}$; K -theory is homotopy invariant, so $\hat{\sigma}_{s*} = \text{id}$ for every s .

(3) The defining property of the dual weight of a KMS_β state is that it is a trace scaled by the dual action, $\hat{\varphi} \circ \hat{\sigma}_s = e^{-\beta s} \hat{\varphi}$; in particular its domain is $\hat{\sigma}$ -invariant. Let p be a projection in the domain. Then $s \mapsto \hat{\sigma}_s(p)$ is a norm-continuous path of projections in the domain, so $\hat{\sigma}_s(p) = u_s p u_s^*$ for a unitary u_s of the unitization, and the trace property $\hat{\varphi}(x^*x) = \hat{\varphi}(xx^*)$ gives $\hat{\varphi}(\hat{\sigma}_s(p)) = \hat{\varphi}(p)$. Hence $\hat{\varphi}(p) = e^{-\beta s} \hat{\varphi}(p)$ for every s , and $\hat{\varphi}(p) = 0$. $\square$

Remark 4.2 (Two mechanisms, one cause). Part (3) is proved here without computing $K_0(\mathcal{B}_{K,S})$, and this is deliberate. By (1), $K_0(\mathcal{B}_{K,S}) \cong K_1(\mathcal{A}_{K,S})$. For $|S| = 1$ this is $\ker(1 - \alpha_*)$ on $C(Y_S, \mathbb{Z})$ by Pimsner’s sequence, and it vanishes: a fixed point of α_* is supported in $\mathfrak{p}^n Y_S$ for every n , hence in the nowhere dense set $\partial Y_S = \bigcap_n \mathfrak{p}^n Y_S$, hence is 0. For $|S| \geq 2$ the group $K_1(\mathcal{A}_{K,S})$ is built from the odd Koszul homology of the commuting family $(1 - \alpha_{\mathfrak{p}*})$, and we have not determined it (Paper II). Were it zero, (3) would follow trivially. The argument above shows that (3) holds regardless: the scaling $\hat{\varphi} \circ \hat{\sigma}_s = e^{-s} \hat{\varphi}$ and the homotopy triviality of $\hat{\sigma}_*$ are incompatible with a nonzero pairing, whatever the group is. It is the *connectedness* of $\mathbb{R}$ that kills the pairing, not the vanishing of a K -group — the same mechanism as in Remark 4.3.

Remark 4.3 (Connected versus totally disconnected). The obstruction in (2) is general: any action of a *connected* group induces the trivial action on K -theory. The modular flow is an action of $\mathbb{R}$, so the scaling structure it carries (namely $\hat{\varphi} \circ \hat{\sigma}_s = e^{-s} \hat{\varphi}$, where the type and the flow of weights live) is invisible to K -theory in principle. By contrast G_S is profinite, hence totally disconnected, and acts nontrivially on $K_0(C(Y_S)) = C(Y_S, \mathbb{Z})$ by permuting clopen sets. Arithmetic symmetry, being Galois-theoretic and so profinite, can register on K -theory; modular dynamics, being a flow, cannot.

Proposition 4.4 (The determinant does not act on K_1). *Let A be unital with a trace τ . The de la Harpe–Skandalis determinant [6] is a homomorphism $\Delta_\tau : \mathcal{U}_0(A) \rightarrow \mathbb{R}/\tau_*(K_0(A))$ on the connected component of the unitary group. Since $K_1(A) = \mathcal{U}_\infty(A)/\mathcal{U}_{\infty,0}(A)$, the domain of Δ_τ is precisely the subgroup by which K_1 is the quotient; Δ_τ descends to $\mathcal{U}_0/\overline{DU}_0$ and does not induce a map $K_1(A) \rightarrow \mathbb{R}$.*

Proof. Definitional: Δ_τ vanishes on commutators of $\mathcal{U}_0$, and a homomorphism defined on $\mathcal{U}_0$ carries no information about $\mathcal{U}/\mathcal{U}_0$. $\square$

Remark 4.5 (Parity). A trace is a cyclic 0-cocycle and pairs with K_0 ; pairing with K_1 requires a cyclic 1-cocycle. On the core the natural candidate is $\psi(b_0, b_1) = \hat{\varphi}(b_0 \hat{\delta}(b_1))$ with $\hat{\delta}$ the generator of the dual flow, but cyclicity needs $\hat{\varphi} \circ \hat{\delta} = 0$ whereas the defining property of the dual weight is $\hat{\varphi} \circ \hat{\sigma}_s = e^{-s} \hat{\varphi}$, i.e. $\hat{\varphi} \circ \hat{\delta} = -\hat{\varphi} \neq 0$. The pair $(\hat{\varphi}, \hat{\delta})$ produces a *twisted* cyclic 1-cocycle, consistently with Theorem 2.3.

Proposition 4.6 (Groupoid homology carries no β). $H_*(\mathcal{G}_S; \mathbb{Z}) \cong H_*(I_S; C_c(\tilde{Y}_S, \mathbb{Z}))$ is defined without reference to β , as is any twist by the modular cocycle $c(\mathbf{a}\mathbf{b}^{-1}) = \log \mathbf{N}\mathbf{a} - \log \mathbf{N}\mathbf{b}$, which does not involve β . Hence neither can determine Ξ_β , which varies with β . Twisting the coefficients instead does not help: with $\mathbb{R}_\beta$ denoting $\mathbb{R}$ with I_S acting by $\mathbf{a} \mapsto \mathbf{N}\mathbf{a}^{-\beta}$,

$$H^n(I_S; \mathbb{R}_\beta) = 0 \quad \text{for all } n \geq 0,$$

while H^0 with coefficients in the measures twisted by $e^{-\beta c}$ returns the KMS simplex tautologically.

Proof. The first statements are immediate: β enters only as the exponent in the Radon–Nikodym derivative, i.e. in the measure class. For the vanishing, pick $\mathbf{p} \in S$; the generator $\mathbf{p}$ acts on $\mathbb{R}_\beta$ by $\lambda = \mathbf{N}\mathbf{p}^{-\beta} \neq 1$, and for $\mathbb{Z}$ acting on $\mathbb{R}$ by $\lambda \neq 1$ one has $H^0 = \ker(\lambda - 1) = 0$ and $H^1 = \mathbb{R}/(\lambda - 1)\mathbb{R} = 0$; the Koszul complex for $I_S = \mathbb{Z}^n$ is then exact, since one of its generators acts by an automorphism $\lambda - 1$ of the coefficients, and for infinite S the Milnor sequence of the increasing union of finite-rank subgroups gives the same vanishing. The last statement is the definition of quasi-invariance together with [13, Prop. 2.8]. $\square$

5. DYNAMICAL ENTROPY

Proposition 5.1 (Scaling, not invariance). For a KMS_β state φ , $\mathbf{a} \in J_S$ and $x \in \mathcal{A}_{K,S}$,

$$\varphi(\alpha_{\mathbf{a}}(x)) = \mathbf{N}\mathbf{a}^{-\beta} \varphi(x), \quad \varphi(\mathcal{L}_{\mathbf{a}}(x)) = \mathbf{N}\mathbf{a}^\beta \varphi(x \mathbf{1}_{\mathbf{a}Y_S}),$$

where $\mathcal{L}_{\mathbf{a}}(x) = \mu_{\mathbf{a}}^* x \mu_{\mathbf{a}}$ is the unital completely positive transfer operator. Neither map preserves φ for $\mathbf{a} \neq (1)$, $\beta > 0$.

Proof. $\sigma_{i\beta}(\mu_{\mathbf{a}}) = \mathbf{N}\mathbf{a}^{-\beta} \mu_{\mathbf{a}}$ and $\sigma_{i\beta}(\mu_{\mathbf{a}}^*) = \mathbf{N}\mathbf{a}^\beta \mu_{\mathbf{a}}^*$. The KMS condition $\varphi(uv) = \varphi(v\sigma_{i\beta}(u))$ with $u = \mu_{\mathbf{a}}$, $v = x\mu_{\mathbf{a}}^*$ gives the first, using $\mu_{\mathbf{a}}^* \mu_{\mathbf{a}} = 1$; with $u = \mu_{\mathbf{a}}^*$, $v = x\mu_{\mathbf{a}}$ it gives the second. For the failure of the second, take $x = \mathbf{1}_E$ with $E = Y_S \setminus \mathbf{a}Y_S$: the right side is 0 while $\varphi(\mathbf{1}_E) = 1 - \mathbf{N}\mathbf{a}^{-\beta} > 0$. $\square$

Theorem 5.2 (No invariant measure, hence no metric entropy). For $S \neq \emptyset$ there is no I_S -invariant Borel probability measure on $\tilde{Y}_S$ other than those concentrated on the finite boundary ∂Y_S , and no such measure is a KMS_β state for $\beta > 0$. Consequently the Connes–Størmer, CNT and Sauvageot–Thouvenot entropies [5, 3, 9] $h_\varphi(\alpha_{\mathbf{a}})$ and $h_\varphi(\mathcal{L}_{\mathbf{a}})$ are undefined for every KMS_β state.

Proof. An invariant ν satisfies $\nu(\mathbf{a}^{-1}Y_S) = \nu(Y_S)$ for all $\mathbf{a} \in J_S$, and $\tilde{Y}_S = \bigcup_{\mathbf{a}} \mathbf{a}^{-1}Y_S$, so $\nu(Y_S) = 1$; then $\nu(\mathbf{p}^k Y_S) = 1$ for all k and ν is concentrated on $\bigcap_k \mathbf{p}^k Y_S$; intersecting over $\mathbf{p} \in S$ concentrates ν on $\partial Y_S = \{[0, g]\} \cong G_S/r(U_S) \cong \text{Cl}_S^+$ [13, (2.1)], a finite set. A KMS_β state has $\varphi(\mathbf{1}_{\mathbf{p}Y_S}) = \mathbf{N}\mathbf{p}^{-\beta} < 1$ and so is not of this form. All three entropy definitions require the state to be invariant under the map whose entropy is computed; Proposition 5.1 shows no KMS state is. $\square$

Remark 5.3 (Consistency with type III). Theorem 5.2 restates the type classification. A type III factor has no semifinite trace, the associated relation has no invariant measure in the measure class, and Kolmogorov–Sinai theory is unavailable by construction.

Theorem 5.4 (Topological entropy vanishes). Let $T_{\mathbf{a}} : Y_S \rightarrow Y_S$, $[x, g] \mapsto [\mathbf{a}x, \sigma_{\mathbf{a}}g]$, so that $\mathcal{L}_{\mathbf{a}}(f) = \mu_{\mathbf{a}}^* f \mu_{\mathbf{a}} = f \circ T_{\mathbf{a}}$ on $C(Y_S)$. Then $h_{\text{top}}(T_{\mathbf{a}}) = 0$. On the quotient of $\mathcal{A}_{K,S}$ in which every

$\mu_{\mathfrak{p}}$ is unitary (the boundary quotient of Paper II) the map $\alpha_{\mathfrak{a}}$ becomes the inner automorphism $\text{Ad } \mu_{\mathfrak{a}}$, whose Voiculescu–Brown topological entropy [11, 2] is 0.

Proof. Y_S is the quotient of $\widehat{\mathcal{O}}_S \times G_S$ by U_S , and $T_{\mathfrak{a}}$ lifts to $\widetilde{T}_{\mathfrak{a}}(x, g) = (\mathfrak{a}x, \sigma_{\mathfrak{a}}g)$. Put

$$d((x, g), (x', g')) = \sup_{\mathfrak{p}} c_{\mathfrak{p}} \text{Np}^{-v_{\mathfrak{p}}(x_{\mathfrak{p}} - x'_{\mathfrak{p}})} + d_G(g, g'),$$

with $c_{\mathfrak{p}}$ summable and d_G an invariant metric on G_S ; one has $v_{\mathfrak{p}}(\mathfrak{a}(x - x')) = v_{\mathfrak{p}}(\mathfrak{a}) + v_{\mathfrak{p}}(x - x')$ and $d_G(\sigma_{\mathfrak{a}}g, \sigma_{\mathfrak{a}}g') = d_G(g, g')$, so $\widetilde{T}_{\mathfrak{a}}$ is 1-Lipschitz. Hence the Bowen metrics satisfy $d_n = d$, the (n, ϵ) -separated sets are the ϵ -separated sets, their cardinality is independent of n , and $h_{\text{top}}(\widetilde{T}_{\mathfrak{a}}) = 0$; a factor of a zero-entropy system has zero entropy. For the last statement, $\text{Ad } \mu_{\mathfrak{a}}$ is inner, and inner automorphisms have zero topological entropy [2]. $\square$

Remark 5.5 (Why $\log \text{Na}$ does not occur). The map with entropy $\log \text{Np}$ is the digit shift $x \mapsto (x - a_0)/\mathfrak{p}$, which is Np -to-one; its definition requires *subtracting* the last digit, i.e. the additive structure. The multiplicative system has one isometry per prime, with range projections $\mathbf{1}_{\mathfrak{p}Y_S}$ that are nested rather than orthogonal ($\mathbf{1}_{\mathfrak{p}Y_S}\mathbf{1}_{\mathfrak{q}Y_S} = \mathbf{1}_{\mathfrak{p}\mathfrak{q}Y_S} \neq 0$), hence no Cuntz relation and no $\log n$ (Paper II). The unavailability of $\log \text{Na}$ is the same phenomenon as the failure of the Cuntz relation there.

Theorem 5.6 (The modular flow has zero entropy). *Let $\iota : \mathbb{R} \rightarrow \widehat{I}_S = \prod_{\mathfrak{p} \in S} \mathbb{T}$, $\iota(t)_{\mathfrak{p}} = \text{Np}^{it}$, and let γ be the gauge action. Then $\sigma_t = \gamma_{\iota(t)}$, so $\{\sigma_t\}$ has compact closure in $\text{Aut}(\mathcal{A}_{K,S})$; and an automorphism with relatively compact orbit closure has zero CNT entropy. Hence $h_{\varphi}(\sigma_t) = 0$ for every KMS $_{\beta}$ state and every t , extremal or symmetric alike.*

Proof. ι is a continuous homomorphism and both σ_t and $\gamma_{\iota(t)}$ fix $C(Y_S)$ pointwise and send $\mu_{\mathfrak{p}} \mapsto \text{Np}^{it}\mu_{\mathfrak{p}}$; the gauge action of the compact group $\widehat{I}_S$ is continuous for the pointwise norm topology, so its image is compact. For the entropy, fix a finite-dimensional channel γ and $\epsilon > 0$; by relative compactness every $\sigma_{kt} \circ \gamma$ is within ϵ of one of finitely many $\theta_j \circ \gamma$, $j \leq m(\epsilon)$. Since H_{φ} is symmetric, unchanged by repetition, subadditive, bounded by $\log \dim$ on a single channel, and continuous with a modulus independent of the number of arguments,

$$H_{\varphi}(\gamma, \sigma_t\gamma, \dots, \sigma_{(n-1)t}\gamma) \leq m(\epsilon) \log \dim + n\delta(\epsilon),$$

so $\limsup_n \frac{1}{n} H_{\varphi} \leq \delta(\epsilon) \rightarrow 0$. Every KMS state is σ -invariant, so the hypothesis holds. The properties of H_{φ} used here are those established in [3]; see also [8] for the vanishing of CNT entropy on automorphisms lying in a compact group. $\square$

Remark 5.7. This is the noncommutative form of the fact that rotations have zero entropy. Here the angle is $(\log \text{Np})_{\mathfrak{p} \in S}$; over $\mathbb{Q}$ these are $\mathbb{Q}$ -linearly independent and Kronecker's theorem makes $\iota(\mathbb{R})$ dense in $\widehat{I}_S$, and in general $\iota(\mathbb{R})$ is dense in the annihilator of $\{\mathfrak{a} \in I_S : \text{Na} = 1\}$. The modular flow is an irrational winding on an infinite torus, dense hence ergodic on gauge orbits, compact hence entropy-free. Neither property detects Ξ_{β} , which by Corollary 2.4 is not an invariant of an automorphism.

6. RELATIVE ENTROPY AND THE INFORMATION STAIRCASE

Fix β with Ξ_{β} finite and write $G = \widehat{\Xi}_{\beta}$, A_{β} for the set of thresholds already crossed in the cascade systems of [13, Prop. 4.16], so that $\Xi_{\beta} \cong (\mathbb{Z}/2\mathbb{Z})^{A_{\beta}}$ and $|G| = 2^{|A_{\beta}|}$.

Theorem 6.1 (The staircase). *Let φ_{ε} , $\varepsilon \in G$, be the extremal KMS $_{\beta}$ states and φ_{sym} the symmetric one. Then*

$$(6.1) \quad S(\omega_{\varphi_{\varepsilon}} \parallel \omega_{\varphi_{\text{sym}}}) = \log |G| = |A_{\beta}| \log 2,$$

computed in $M_{\text{sym}} = \pi_{\varphi_{\text{sym}}}(\mathcal{A}_{K,S})'' = \bigoplus_{\varepsilon} M_{\varepsilon}$, and the whole value is carried by the centre: each type III summand contributes zero. For $\varepsilon \neq \varepsilon'$, $S(\omega_{\varphi_{\varepsilon}} \parallel \omega_{\varphi_{\varepsilon'}}) = +\infty$. When A_{β} is infinite the two states are mutually singular and (6.1) is $+\infty$.

Proof. By [13, Thm. 3.6] the extreme points of the KMS_{β} simplex are indexed by G and their measures are ergodic for the orbit relation and pairwise distinct, hence mutually singular; so the GNS representations are pairwise disjoint, $\varphi_{\text{sym}} = |G|^{-1} \sum_{\varepsilon} \varphi_{\varepsilon}$ gives $\pi_{\varphi_{\text{sym}}} = \bigoplus_{\varepsilon} \pi_{\varphi_{\varepsilon}}$ and the central decomposition $M_{\text{sym}} = \bigoplus_{\varepsilon} M_{\varepsilon}$, with central projections z_{ε} of $\omega_{\varphi_{\text{sym}}}$ -value $|G|^{-1}$. For a direct sum with $\omega = \bigoplus \lambda_i \omega_i$ and $\psi = \bigoplus \mu_i \psi_i$ one has the additivity formula

$$S(\omega \parallel \psi) = \sum_i \lambda_i \log \frac{\lambda_i}{\mu_i} + \sum_i \lambda_i S(\omega_i \parallel \psi_i),$$

with the conventions $0 \log 0 = 0$ and $\lambda \log(\lambda/0) = +\infty$ for $\lambda > 0$; this is the relative modular operator of a direct sum decomposing as the direct sum of the relative modular operators scaled by λ_i/μ_i [1], [8]. Index the summands by $\eta \in G$. The state $\omega_{\varphi_{\varepsilon}}$ has $\lambda_{\eta} = 1$ for $\eta = \varepsilon$ and $\lambda_{\eta} = 0$ otherwise, while $\omega_{\varphi_{\text{sym}}}$ has $\mu_{\eta} = |G|^{-1}$ for every η . The single surviving term of the first sum is $\log(1/|G|^{-1}) = \log |G|$. In the second sum the term $\eta = \varepsilon$ vanishes because both states induce the *same* normal state of M_{ε} , and every other term has $\lambda_{\eta} = 0$; so the second sum is identically zero, which is the assertion that the relative entropy is purely central. Disjointness gives $+\infty$ for $\varepsilon \neq \varepsilon'$. When A_{β} is infinite, $\Xi_{\beta}^{\perp}$ has infinite index in G_S , the extremal measure is Haar measure on a coset of $\Xi_{\beta}^{\perp}$ in each fibre while the symmetric one is Haar measure on G_S , so the two are mutually singular and the relative entropy is $+\infty$. $\square$

Corollary 6.2 (One bit per threshold). *Each threshold contributes exactly $\log 2$ to (6.1), and the function $\beta \mapsto |A_{\beta}| \log 2$ is a pure jump staircase with jumps $\log 2$ at the thresholds. It coincides with $\log[M_{\text{sym}} : M_{\varepsilon}]$, the logarithm of the index of the symmetry-breaking inclusion $M_{\varepsilon} \cong M_{\text{sym}}^G \subset M_{\text{sym}} \cong M_{\varepsilon} \otimes \ell^{\infty}(G)$, which is $|G|$ (Paper II of the series): the index staircase and the information staircase are the same object, both counting log of the number of phases.*

Remark 6.3 (The order parameter). For the single-threshold system of [13, Prop. 4.12] the two phases are separated by the pointwise invariant Ψ , whose almost-everywhere definedness is a Borel–Cantelli condition. Under φ_{sym} the observable Ψ takes the values ± 1 with probability $\frac{1}{2}$ each, so $H_{\varphi_{\text{sym}}}(\Psi) = \log 2$, while $H_{\varphi_{\pm}}(\Psi) = 0$: crossing the threshold freezes one previously maximally mixed bit. This is (6.1) in its elementary form.

Proposition 6.4 (No preferred phase). *At fixed β the KMS_{β} states form, by [13, Thm. 3.6], the simplex $\text{Prob}(G_S/\Xi_{\beta}^{\perp})$, on whose extreme points G_S acts transitively. Every extremal state is therefore carried to every other by a symmetry of the dynamical system, and φ_{sym} is the barycentre of the G_S -orbit of any of them. No quantity attached to the pair $(\mathcal{A}_{K,S}, \sigma)$ distinguishes one extremal KMS_{β} state from another, and none singles out φ_{sym} except its symmetry.*

Proof. The action of G_S on the simplex factors through $G_S/\Xi_{\beta}^{\perp}$ and permutes the extreme points simply transitively [13, Thm. 3.6(3)]; the symmetries commute with σ_t . $\square$

Remark 6.5 (What the bit is). Landauer’s bound concerns *erasure*: an external agent irreversibly resetting a bit, which requires work $k_B T \log 2$. Here nothing is erased and no agent acts. Crossing a threshold makes an observable well defined almost everywhere, and the system then has two equilibrium phases in which it takes a definite value. The bit is not destroyed but *transferred*, from the microscopic description to the label of the phase; the $\log 2$ of (6.1) is the information needed to say which phase one is in, not information lost. By Proposition 6.4 no phase is energetically preferred, so the analogy with Landauer does not survive. (In the Gibbs regime $\zeta_{K,S}(\beta) < \infty$ one can say more: the extremal states are the normalized Gibbs states of [13, Thm. 3.6], all with the

same partition function, hence literally the same free energy; in the type III regime no free energy is defined, and the symmetry argument is all there is.)

7. ASSESSMENT

Candidate invariant	Sees Ξ_β ?	Reason
$K_*(\mathcal{A}_{K,S})$	no	Cor. 3.3: coinvariants kill the eigenvectors
$K_*(\mathcal{B}_{K,S})$	no	Thm. 4.1(1)
dual trace on $K_0(\mathcal{B}_{K,S})$	no	Thm. 4.1(3): scaling vs. homotopy
$\hat{\sigma}_*$ on $K_*(\mathcal{B}_{K,S})$	no	Thm. 4.1(2): $\mathbb{R}$ connected
$\Delta_{\hat{\varphi}}$ on K_1	not defined	Prop. 4.4: domain is $\mathcal{U}_0$
$H_*(\mathcal{G}_S; \mathbb{Z})$ and c -twists	no	Prop. 4.6: no β
$h_\varphi(\alpha_a), h_\varphi(\mathcal{L}_a)$	not defined	Thm. 5.2: no invariant measure
$ht(\alpha_a)$	0	Thm. 5.4: a contraction
$h_\varphi(\sigma_t)$	0	Thm. 5.6: compact gauge subflow
$(K_0(C(Y_S)), \alpha_*)$	yes	Thm. 3.2
$H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$	yes	Thm. 2.3
$S(\omega_{\varphi_\varepsilon} \parallel \omega_{\varphi_{\text{sym}}})$	yes	Thm. 6.1

The three affirmative entries have a common form: each is built directly from a Radon–Nikodym derivative. The eigenvalue equation of Theorem 3.2 is the statement that $d(\nu \circ \alpha_p)/d\nu = Np^{-\beta}$; the cohomology of Theorem 2.3 is computed with respect to the measure class of π_β , that is, modulo the Radon–Nikodym cocycle; and the relative entropy of Theorem 6.1 is by definition $\int \log(d\omega/d\psi) d\omega$. Each therefore compares two objects in the same measure class, rather than iterating a single map or extracting a homotopy invariant of a single algebra. The negative entries are exactly the constructions that discard the derivative: a C^* -algebra retains no measure, a topological groupoid retains no measure class, and a homotopy functor cannot see a scaling. That is the content of Corollary 2.4, and it is the reason the negative entries are uniform.

Three questions remain. We have not computed $H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$ itself, only its kernel; for the tail relation of a product measure this should be accessible, and would presumably exhibit $\widehat{G_S}/\Xi_\beta$ inside it. Second, Lemma 2.1 is stated for $\mathbb{T}$ -valued cocycles; the extended group of [12], defined by cocycles valued in $G_S \times \mathbb{R}$, is the corresponding kernel for the factor type, and there the inclusion $E(\Phi) \subseteq \Lambda^\perp$ of Lemma 2.1 yields only an upper bound on the type; whether the kernel determines the type — i.e. whether the cocycle is regular, or equivalently whether the flow of weights has pure point spectrum — is open (Remark 2.2). Third, the twisted cyclic 1-cocycle of Remark 4.5 is the natural home for a pairing with $K_1(\mathcal{B}_{K,S})$; whether the resulting invariant recovers Theorem 3.2 we do not know.

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